standard protocol for very light devices in very weak networks.

CoAP is very easy to understand. It is similar to HTTP, but smaller in size.

- It employs the same concept of resources in URLs as *coap://sensor1/temperature.*
- It works with the same methods as GET, POST, PUT and DELETE.
- It is a non-blocking protocol (UDP) rather than a blocking protocol (TCP), reducing overhead.
- It has a 4-byte fixed header.

There's an *Observe* option too in CoAP. A client may request that it be informed when the value changes. This prevents the device from repeating the same reading over and over again.

CoAP is an interoperability enabler as it can be mapped to HTTP. With a simple gateway, it is possible to convert CoAP requests to HTTP requests. So the tinier battery sensor can become a part of a regular web system.

Open source support: libcoap (C), Eclipse Californium (Java) and aiocoap (Python) are popular implementations. CoAP is also added in the embedded operating systems included in Zephyr, RIOT and Contiki-NG.

OPC UA: The language of the factory

OPC UA is an acronym for Open Platform Communications Unified Architecture. It is managed by the OPC Foundation and has been published as international standard IEC 62541. It is extensively used in factories and industrial automation.

MQTT and CoAP mainly solve the syntactic level of the problem. They carry messages but they don't convey the meaning of the data. OPC UA does more. It also resolves the semantic level of the challenge.

OPC UA is based on an information model. Each machine has a structured description stating what it is, what values it has, the units, and the relationship between the units. For instance, a pump has a speed, a pressure and a temperature, all of which have a name, type, and a unit.

This allows the access of a new client to a machine and the browsing. The client can find out the data without the need to read a vendor handbook. Companion specifications for OPC UA have been developed by industry groups as well. They are the common models for robots, machine tools, packaging machines, and so on.

OPC UA can be used for encryption, signing and certificate-based authentication. Recent versions include a publish/subscribe mode that can run on top of MQTT.

Open source support: There are several popular implementations of open source, such as open62541 (C), Eclipse Milo (Java) and the Python package asyncua.

These three standards are not competitive (see Table 1 for a comparison between them) and are best used in conjunction. Each is used in a system where it is most appropriate and connected by an edge gateway. This is illustrated in Figure 2.

The gateway is the translator. It can read CoAP, OPC UA or even old Modbus data, which it translates to a single format and broadcasts to an MQTT broker. Then apps subscribe to the topics of interest. It is much easier to build such gateways with open source tools such as Node-RED and EdgeX Foundry.

Other open standards

MQTT, CoAP and OPC UA are not alone. Some additional open standards fill crucial gaps.

LwM2M (Lightweight M2M): This standard has been developed by OMA SpecWorks on top of CoAP. It provides a standardised approach in remote administration of devices -- read settings, update firmware, check health.

Sensor Measurement Lists (SenML): Defined in RFC 8428, it has a simple sensor reading format – either JSON or CBOR. It establishes names and units, with '25 Cel' being the same temperature in every location.

Matter: This is an initiative by Connectivity Standards Alliance and was released in 2022. It enables intelligent devices from Apple, Google, Amazon, Samsung and others to communicate with each other. It is open source with a reference code.

AMQP: This offers reliable business messaging via an OASIS and ISO standard. It's frequently employed in the IoT gateway to enterprise system interface.

Devices that can work together are key to the functioning of IoT. Closed systems produce islands. Open standards are the bridges that unite them.

MQTT provides a lightweight and easy way to transport data. CoAP enables access to the smallest devices via the web. OPC UA is an additional layer of meaning and security for industrial machines. Together, they create a solid open stack in conjunction with gateways and common data models.

This stack is free and easy to adopt thanks to open source. The team, small or large, can develop systems that connect using tools such as Mosquitto, Paho, libcoap and open62541. The message for students, startups, and industry is clear. Communicate using open standards and your IoT system will be able to speak with the world. END

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